

Supplementary Material — Knowing When to Stop Asking: A Formal Coverage Criterion for Multi-Turn User Intent Understanding

Anonymous submission

How to read this document with the code repository

This supplement accompanies the main paper. The code, data and per-run records it refers to are released, anonymised for review, at

<https://github.com/aonymous412/interview-pipeline-release>

and this document is `supplement.pdf` in the same repository. The table below maps each section to the files and directories of the repository it draws on (each path is a link); the sections that rest on no released file say so. `python verify_numbers.py` at the repository root recomputes every headline figure of the paper from the stored records, and each section below closes with the exact command or record that reproduces its own numbers.

Section	Repository files and directories
§ A Reproducibility Artefacts	code/phase1/run_multiturn.py code/phase1/run_multiturn_claude.py code/phase1/scoring.py code/ali/user_simulator.py code/ali/components/ data/ results/ verify_numbers.py
§ B Baseline Interview Protocol	code/phase1/ results/baselines/
§ C Per-Domain Results	results/ali/ code/ali/per_domain_baselines.py code/ali/run_theta_sensitivity.py verify_numbers.py
§ D Cost and Latency per Interview	results/ali/ code/ali/cost_analysis.py code/ali/cost_instrumentation.py code/ali/run_cost_latency.py
§ E Deployment Details and Walkthroughs	results/ali/results_ALI_v1.json data/taxonomies.json training/ (record P12 of the first file)
§ F Question Selection: Thresholds, Reward Weights and Training Pipeline	training/ code/ali/ali_pipeline.py code/ali/config.py code/ali/components/c3_generator.py code/ali/ablation.py code/ali/ablation_stats.py reports/ablation_stats.json
§ G Computing Infrastructure	requirements.txt
§ H Real-User Validation Protocol	<i>no file: the protocol has not run</i>

Preliminary work. Under review for the European AI Summer Research Program. Do not distribute.

Section	Repository files and directories (continued)
§I Theoretical Analysis of the Coverage Criterion	<i>no file</i>
§J Positioning Against Prior Work	<i>no file</i>
§K Limitations	<i>no file</i>
§L Ethics Statement	<i>no file</i>

A Reproducibility Artefacts

This section gives what a reader needs to re-derive every coverage figure in the paper without re-running a model: the four prompts that define the protocol, verbatim from the code; the stored artefacts and the recomputation they allow; and a reproducibility checklist.

A.1 The Four Prompts, Verbatim

The prompt of a baseline is part of the protocol, and the asymmetries discussed in Section 5 of the main paper are visible in the text below rather than in any number. All six multi-turn baselines of Table 2 of the main paper share the interviewer and oracle prompts; `{domain_task}` is replaced at run time by “scope and build a software project”, “generate a complete clinical oncology report” or “configure an autonomous financial trading/payment agent safely”.

Baseline interviewer (system prompt).

You are an expert requirements interviewer. Your task is to gather the information needed to `{domain_task}` from the user.

MANDATORY RULES -- FOLLOW EXACTLY:

1. Ask EXACTLY ONE focused question per message -- never a list of questions
2. Do NOT ask for information already provided by the user
3. Prioritize the most critical missing information first
4. YOU MUST OUTPUT [DONE] AFTER 5 TO 8 QUESTIONS -- this is not optional
5. When you have gathered sufficient key requirements, write [DONE] on its own line, exactly like this:

[DONE]

6. After [DONE], write a short bullet-point summary of what was gathered

CRITICAL: You are required to stop and output [DONE] after 5-8 questions maximum.

Do NOT ask more than 8 questions under any circumstances. Quality over quantity.

Baseline simulated user (system prompt). Rule 4 lets the baseline’s user decline a concrete value, which the scorer never credits.

You are a user who needs to `{domain_task}`. Your original request was:

`"{original_prompt}"`

RULES:

1. Answer the interviewer’s questions based on what a realistic user with this project/situation would know
2. Give natural, concise answers (1-3 sentences)
3. If asked about something you haven’t thought about, give a reasonable answer
4. If asked something very technical, say "I’m not sure" or give a vague preference
5. Stay consistent with previous answers
6. Don’t volunteer extra information beyond what was asked
7. Be realistic -- not every user knows exact budgets, timelines, or technical details

Harness reminders appended to the simulated user’s reply. The first is appended from turn five, the second from turn eight ({turn} is the running count).

```
[REMINDER: You have asked {turn} questions. Wrap up and output [DONE] soon.]
[REMINDER: You have asked {turn} questions. You MUST output [DONE] now -- do not
ask another question.]
```

ALI simulated user (system prompt). ALI receives no instruction about when to stop; it halts on $\text{cov}(F) \geq \theta_d$.

```
You are a user with the following need: {initial_prompt}
You have all the information needed to answer questions but only reveal it when
directly asked. Answer concisely and naturally. If asked for something not
specified in your need, invent a plausible concrete value consistent with it.
Never volunteer information that wasn’t explicitly asked.
```

Scorer (LLM-as-judge, one call per completed transcript, temperature 0). Only RESOLVED contributes weight. The parameter list, with weights and descriptions, is the domain’s entry of `taxonomies.json`; the response is truncated to 5,000 characters.

```
You are a precise annotation assistant for a research project.
TASK: Classify each parameter as RESOLVED, MENTIONED, or ABSENT.
DEFINITIONS:
- RESOLVED: A CONCRETE value is given (e.g., "$5,000", "women aged 25-40", "stage
IIB")
- MENTIONED: Discussed but no concrete value committed
- ABSENT: Not addressed at all
DOMAIN: {domain}
ORIGINAL PROMPT: {prompt_text}
PARAMETERS:
{param_descriptions}
RESPONSE TO SCORE:
--
{response_text[:5000]}
--
Return ONLY a JSON object. Keep evidence very short (max 10 words).
Example: {"param_name": {"status": "RESOLVED", "evidence": "value found"}}
```

A.2 Stored Artefacts and What They Allow

- **Prompt set.** The 150 evaluation prompts, 50 per domain, 43 ultra-vague, 62 medium and 45 detailed, in `prompts.json` (id, domain, vagueness tier, text).
- **Parameter spaces.** The three weighted taxonomies with their descriptions; the scorer and the pipeline read the same file, `taxonomies.json`.
- **One scored artefact per configuration.** For every run: the resolved frame F , the transcript, coverage after each turn and the termination reason. Coverage recomputes as $\sum_{p \in F} w_d(p) / \sum_{p \in \mathcal{P}_d} w_d(p)$ without any model call; recomputing all 150 ALI_v1 runs returns the stored value exactly.
- **Scorer validation sheet.** The 516 double-annotated items (scorer label and human label side by side), so $\kappa = 0.749$ overall and $0.758 / 0.968 / 0.539$ per domain recompute, including against us.

- **Bootstrap and tests.** Percentile bootstrap, $B = 10,000$, seed 42; paired Wilcoxon signed-rank with Holm–Bonferroni and TOST margins fixed before the ablation ran.
- **Run-to-run resolution.** Re-executing an unchanged configuration two weeks apart moves a domain mean by 0.9 pp, with 37 of 50 runs identical.
- **Termination audit.** Of 500 multi-turn baseline runs, 499 end on the interviewer’s declaration and the maximum observed length is nine turns; ALI_v1 exceeds eight turns on 27 of 50 Consulting, 25 of 50 Medical and 0 of 50 Payments runs. The eight-turn truncation of Table 2 of the main paper is a re-analysis of the stored ALI_v1 trajectories, not a new run.

A.3 Code and Data Availability

Code, prompts, taxonomies, stored runs and the annotation sheet are available at <https://github.com/aonymous412/interview-pipeline-release> (anonymised for review).

A.4 Reproducibility Checklist

Item	Status
Conceptual outline and pseudocode of the method	yes
Opinions and hypotheses delineated from results	yes
Assumptions of the theoretical analysis stated formally; proofs included	yes
Theoretical claims demonstrated empirically	partial
Motivation for the selected datasets	yes
Novel datasets (prompt set, taxonomies) included and released for research use	yes
Existing datasets cited; publicly available	yes / partial
Number and range of hyper-parameter values tried, with selection criterion	partial
Pre-processing code included	partial
Source code for running and analysing the experiments included; released for research use	partial / yes
Seeding described sufficiently for replication	partial
Computing infrastructure specified	yes
Evaluation metrics formally described and motivated	yes
Number of runs per reported result stated	yes
Measures of variation or confidence reported	yes
Statistical tests used for claimed differences	yes
Final hyper-parameters listed for each model	partial

Reproduce: `python verify_numbers.py` recomputes every coverage figure of the paper from the stored records, opening with the integrity pass described above; `python code/ali/evaluate.py` gives the per-configuration means and bootstrap intervals.

B Baseline Interview Protocol

Bounds contributions (4) and (5). The multi-turn baselines did not run the same race as ALI. Section 5 of the main paper states the three asymmetries; this section gives the operative instructions and the resulting termination behaviour, so a reader can judge their size rather than take it on trust. All six multi-turn baselines share one system prompt, which the frontier runner imports from the Phase 1 script and which we ship unmodified in the code and data release as the constant `INTERVIEWER_SYSTEM` of `code/phase1/run_multiturn.py`. It casts the model as “an expert requirements interviewer” gathering what is needed to `{domain_task}`, which the runner replaces at run time by “scope and build a software project”, “generate a complete clinical oncology report” or “configure an autonomous financial trading/payment agent safely”. Six rules follow, under `MANDATORY RULES — FOLLOW EXACTLY`: three ordinary interviewing instructions (one focused question per message, never re-ask, take the most critical gap first) and two output-protocol rules (`[DONE]` on its own line once the model has gathered enough, then a bullet summary). The fourth rule bounds the comparison, and the prompt states it twice: as rule four, “`YOU MUST OUTPUT [DONE] AFTER 5 TO 8 QUESTIONS -- this is not optional`”, and again in a closing

CRITICAL clause, “You are required to stop and output [DONE] after 5-8 questions maximum. Do NOT ask more than 8 questions under any circumstances. Quality over quantity.” We give ALI no instruction of any kind about when to stop: it halts on $\text{cov}(F) \geq \theta_d$.

The harness reinforces the cap twice: from turn five it appends [REMINDER: You have asked n questions. Wrap up and output [DONE] soon.] to the simulated user’s reply, and from turn eight [REMINDER: ...You MUST output [DONE] now -- do not ask another question.]

What the baselines did. The baselines obeyed the instruction. Of 500 multi-turn baseline runs, 499 end with the interviewer declaring it has heard enough (the exception is an API error), and the maximum observed anywhere is nine turns, once, by Gemini 2.5 Flash; none approaches the 15-turn cap both harnesses share, and mean turns per system are the *Calls* column of table D.1. We elicited that declaration rather than observing it unprompted, so these runs measure stopping at the instructed budget and their coverage is a lower bound on what the same models would reach unconstrained. ALI, given the same cap and no instruction, exceeds eight turns on 27 of 50 Consulting and 25 of 50 Medical runs, and on none of the 50 Payments runs, where $\theta = 0.75$ fires early. That ordering matches the per-domain margins of Section C, which is why we treat the turn budget as the asymmetry most likely to be doing work.

What this bounds. We measure one asymmetry. Truncating every ALI_v1 run at eight turns and rescoring the partial frame drops overall coverage from 89.0 to 84.4% and ALI’s mean turn count from 6.93 to 5.95, below every frontier baseline, while ALI still leads in each domain (row *8-turn cap* of table C.1), which bounds it at 4.6 pp of the 30.6 pp gap. We do not measure the other two: settling them needs a baseline re-run without the eight-question instruction and with the ALI oracle, and a condition placing the domain taxonomy in the baseline’s own prompt, neither of which this paper contains. Until then the reported gap is an upper bound on what the stopping rule alone contributes, and the flat capability curve is measured at a fixed turn budget.

Reproduce: the eight-turn truncation is the section *8-turn truncation control* of `python verify_numbers.py`; the baseline records are in `results/baselines/`.

C Per-Domain Results

Table C.1: Weighted coverage (%) and turns per domain: same 50 prompts per domain, same scorer, same bootstrap procedure. A domain-specific agent built on the ALI decomposition leads in every domain, and the strongest generalist changes with the domain. $n = 50$ per domain for Claude Opus 4.8, Claude Haiku 4.5 and ALI_v1; 15–20 for `gemma3:4b`, whose per-domain figures are indicative. Strongest baseline measured on a full 50 underlined.

System	Consulting		Medical		Payments	
	Cov.	Turns	Cov.	Turns	Cov.	Turns
<i>General-purpose LLM interviewers, no formal stopping criterion</i>						
<code>gemma3:4b</code>	41.6	4.10	50.5	4.27	34.8	4.20
Claude Haiku 4.5	52.1	6.58	<u>63.5</u>	6.22	55.6	6.50
Claude Opus 4.8	<u>57.3</u>	6.94	54.3	6.42	<u>63.6</u>	6.88
<i>Domain-specific agents on the ALI decomposition</i>						
θ_d		0.90		0.90		0.75
ALI_v1	92.7	8.52	90.6	8.28	83.9	3.98
<i>ALI_v1</i> , 8-turn cap	87.4	7.40	81.9	6.48	83.9	3.98

Table C.1 makes the comparison one domain at a time. Against the strongest baseline measured on the full 50 prompts of each domain, the margin is +35.4 pp in Consulting (92.7 [92.2–93.1] against Claude Opus 4.8, 57.3 [54.5–60.1]), +27.0 pp in Medical (90.6 [89.0–91.7] against Claude Haiku 4.5, 63.5 [58.5–68.6]) and +20.4 pp in Payments (83.9 [82.8–85.0] against Claude Opus 4.8, 63.6 [58.1–69.0]); the confidence intervals are disjoint in all three. Held to the baselines’ eight-turn

budget (row *8-turn cap*), the margins are +30.1, +18.4 and +20.3 pp, still with disjoint intervals. Two features of the table matter more than the margins.

The strongest generalist *changes with the domain* (Opus in Consulting and Payments, Haiku in Medical), as one expects if, at this budget, capability is not the binding constraint. The frontier baselines also spend near-identical effort everywhere (Opus 6.94/6.42/6.88 turns; Haiku 6.58/6.22/6.50) whereas ALI ranges from 3.98 to 8.52, because we set θ_d and the weight vector per domain.

Each ALI row is bounded by its own threshold, and the three do not rank the deployments against one another: Payments halts after 3.98 turns because $\theta_d = 0.75$ and five of its eleven parameters carry 81% of the weight, so the criterion is met early by design (Section 3.4 of the main paper); Payments is also the domain where the scorer agrees least with human annotators ($\kappa = 0.539$). Medical accounts for all four runs that reach the turn cap, and its figure comes from a generic LLM behind C1/C4 rather than the clinical model.

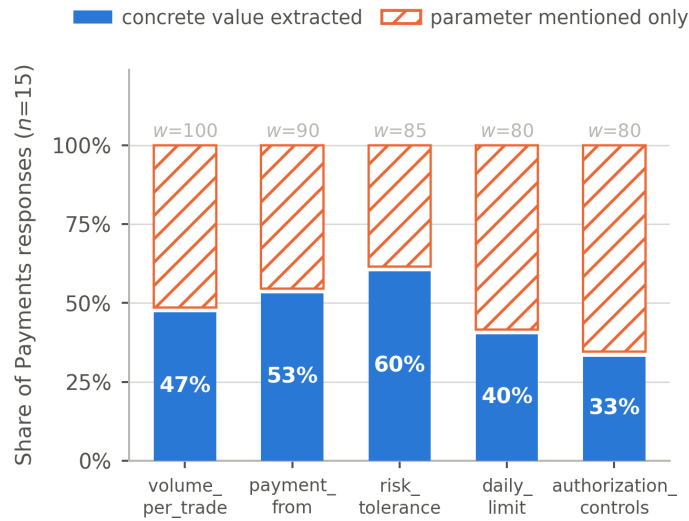


Figure C.1: What coverage counts. Single-pass Gemini 2.5 Flash names the five highest-weight Payments parameters far more often than it resolves them (Section 4.3 of the main paper).

Threshold sensitivity. Per-domain coverage is bounded by its own stopping threshold: the loop halts as soon as $\text{cov}(F) \geq \theta_d$, so a domain with a lower θ_d reports a lower figure by construction. To separate the threshold from the deployment, we re-ran ALI_v1 on the same 50 Payments prompts with the same pipeline, scorer and simulated user, changing only θ from 0.75 to 0.90. Coverage rises from 83.9 [82.8–85.0] to 93.9 [93.4–94.5] and turns from 3.98 to 5.32; the paired gain is +10.0 pp [+8.7, +11.3] and all fifty episodes improve (Wilcoxon signed-rank, $p < 10^{-9}$). At a threshold matching the other two deployments, Payments thus reaches comparable coverage in fewer turns, and within the turn budget the baselines use: only two of the fifty runs exceed eight turns. This is a sensitivity analysis rather than a reconfiguration proposal ($\theta = 0.75$ is the deployed setting), and it does not make the three domains comparable, their parameter spaces differing in size and shape. It does show that the spread between the rows above is a property of θ_d , not evidence that the Payments deployment is weaker. The deployed $\theta = 0.75$ reflects what FinAgent asks of the interview: it elicits a constraint frame once, at agent setup, and the deterministic engine gates every subsequent action, so the marginal value of the last points of coverage is lower than where the frame is the final artefact. In the deployment’s own trading taxonomy, *volume_per_trade* ($w = 35$), *payment_from* ($w = 25$) and *currency* ($w = 20$) yield $\text{cov} = 0.80$ on their own. A uniform $\theta = 0.90$ would raise overall coverage from 89.0% to 92.1% (Table 2 of the main paper).

Reproduce: the per-domain figures are the section *ALI_v1 by domain* of

python verify_numbers.py; python code/ali/per_domain_baselines.py gives the per-domain decomposition; the Payments $\theta = 0.90$ re-run is the record results/ali/results_ALI_v1_payments_theta090.json.

D Cost and Latency per Interview

Table 2 of the main paper reports calls and total tokens per system. This section gives the full breakdown, states how each number was obtained and records the reservations that bound it. Throughout, the simulated user is measured but *excluded* from every system total: it is a shared experimental artefact, not a cost a deployment would pay.

How the numbers were obtained. For the ALI configurations and for single-pass extraction, every model call was instrumented directly: token counts come from the provider’s own usage metadata, and wall-clock time is measured per call. These four rows are measured on the same paired 30-prompt subset used for the ablation. For the six multi-turn baselines no token counter was recorded at the time, so payloads were reconstructed offline from the saved transcripts, which preserve the full conversation resent at each turn. The reconstruction was validated against the real API counters available for the Claude runs: on the four cases without adaptive reasoning it is faithful up to a constant factor of 1.149, with two independent runs agreeing to three decimals. That residual is a tokenizer difference plus per-message API overhead, not a modelling error. The main paper prices systems in tokens; the character column below is kept as the tokenizer-free control, exact and strictly comparable across families, and bounds what the reconstruction can understate. Restricting Claude Opus 4.8 and Claude Haiku 4.5 to the same 30 prompts changes their cost by 1.2% and 1.7%, so mixing sample sizes across rows is not a material distortion.

Table D.1: Cost per completed interview, simulated user excluded. “Formal stop” is the share of runs halted by $\text{cov}(F) \geq \theta_d$ rather than by the turn cap, and is undefined where no such criterion exists. It is computed over the full 150-run evaluation, not over the $n = 30$ cost subset. Token counts for the measured rows come from provider usage metadata; those for the reconstructed rows use a single shared tokenizer and run $\approx 12\text{--}20\%$ below the provider’s own count.

System	n	Source	Calls	Tok. in	Tok. out	Chars	Formal stop
Single-pass Gemini 2.5 Flash	30	measured	1.00	115	3,754	15,658	—
gemma3:4b	50	reconstructed	4.18	1,728	272	9,070	—
llama3.1:8b	50	reconstructed	5.90	2,706	295	14,534	—
Gemini 2.5 Flash	50	reconstructed	7.02	3,283	324	16,830	—
Claude Sonnet 5	50	reconstructed	6.86	4,136	537	21,376	—
Claude Haiku 4.5	150	reconstructed	6.43	4,320	597	22,782	—
Claude Opus 4.8	150	reconstructed	6.75	4,492	713	23,880	—
ALI_v1	30	measured	19.90	5,232	2,575	29,968	97.3%
ALI_v2	30	measured	20.60	7,437	1,966	36,717	96.7%
ALI_v3	30	measured	36.80	10,615	4,965	64,681	30.0%

Where ALI spends. ALI issues $1 + 3n$ calls for n turns: C0 once, then C1, C3 and C4 at every turn; C2 clusters locally and makes no model call at all. Per interview, ALI_v1 spends 462 tokens in C0, 2,196 in C1, 2,213 in C3 and 2,937 in C4. The comparison with a baseline is therefore not one of scale but of shape: a baseline makes one call per turn and resends the entire conversation, so its input grows quadratically and dominates its cost (4,492 input tokens against 713 output for Claude Opus 4.8); ALI makes several short, single-parameter calls whose context stays bounded. This is what lets ALI process only 1.25 times the text of Opus while asking 2.95 times as many questions of a model. ALI_v3 is the counterexample that makes the point: with C4 fine-tuned it needs 64,681 characters and 36.8 calls to reach 51.6% coverage, the worst outcome in the table at more than twice the cost of ALI_v1.

Latency, and why it should be read with care. Wall-clock time per interview is 104.0 s for ALI_v1, 76.4 s for ALI_v2 and 610.1 s for ALI_v3, against 43.9–53.1 s for the Claude baselines.

Three qualifications matter more than the figures themselves. First, the baseline timings are derived from run-stage timestamps and include one scorer call and process start-up per prompt, so they are an *upper* bound: the bias favours ALI. Second, ALI_v1’s slowest component is C1 in local CPU inference (46.9s of 104.0s, averaged over three deployments: 33.7s for the GPT-2 124 M of Telos, 42.5s for the hosted C1 of Rad-Assist and 64.6s for the Qwen2.5-0.5B; the smallest model is not the slowest); ALI_v2, which places C1 behind the same hosted model as the rest, is *faster* at 76.4s despite consuming more tokens. Latency here is a property of the testbed, not of the architecture. Third, wall-clock time depends on API load and hardware and is not reproducible in the way the call and token counts are.

Consistency checks. The instrumented run reproduces the uninstrumented one: ALI_v1 measured here reaches 90.2% coverage in 6.30 turns, against 90.3% and 6.33 turns for the full-pipeline arm of the ablation on the same prompts. ALI_v3 reaches 51.2% here against 51.6% on the full 150-prompt set, with 20 of 30 episodes stopping at the turn cap, the same failure mode reported in the main paper.

Reproduce: characters, tokens and call counts recompute offline with `python code/ali/cost_analysis.py` from the stored transcripts; wall-clock latency does not (it needs the instrumented re-run, `code/ali/run_cost_latency.py`, with API access).

E Deployment Details and Walkthroughs

Table E.1: Architecture comparison across the three case studies. The five-component decomposition is identical in all three; adapters, LoRA configurations and training set sizes are deployment choices and are given below.

	Rad-Assist	Telos	FinAgent
Domain	Medical	Consulting	Payments
Domain count	1 (multi-subtype)	15 project categories	5 (trading, payroll, procurement, moderation, generic)
C1 model	Mistral 7B	GPT-2, 124 M	Qwen2.5-0.5B-Instruct
C3/C4 model	Mistral 7B	GPT-2, 124 M	Qwen2.5-1.5B-Instruct
θ_d	0.90	0.90	0.75
Enforcement	Advisory	Advisory	Deterministic engine

E.1 LoRA Configuration (Telos)

The configuration below applies to Telos only. Rad-Assist does *not* share it: it is built on Mistral 7B trained with GRPO and exported with merged weights, so no separate adapter exists, and in the evaluated pipeline both its C1 and C4 stages fall back to the general-purpose API model.

Table E.2: ALI configuration for Telos (GPT-2 124 M). Rad-Assist uses Mistral 7B+GRPO; see Section 3.4 of the main paper.

Parameter	Value
Base model	GPT-2, 124 M parameters
Fine-tuning	LoRA via PEFT: $r = 16$, $\alpha = 32$, dropout = 0.05
Target modules	<code>c_attn</code> , <code>c_proj</code>
Trainable parameters	1.62 M (1.30% of 124 M)
Training examples	974 (C1), 907 (C4)
Training time / component	$\approx$ 15 minutes (CPU)
θ (evaluated)	0.90
θ (deployed)	0.85–0.95, scaled by domain

E.2 Telos Walkthrough: The Trace Behind Table 1 of the Main Paper

Table E.3: Full trace behind Table 1 of the main paper. Prompt P12: “I want an e-commerce site for a French bakery. It should support online ordering with delivery and pickup options, a loyalty points system, and Stripe for payments.” Answers come from the simulated user and are abridged; weights w are those of the Consulting parameter space ($\sum w = 978$, $\theta = 0.90$).

t	Question (C3)	Answer (simulated user, abridged)	Resolved by C4 (w)	cov
1	What are the main pages or sections you envision for the site?	Home, Shop/Menu, Product Detail Pages, Cart, Checkout, My Account (for order history and loyalty points), Loyalty Program, About Us, Contact Us, Delivery & Pickup Information	pages_structure (90) + user_accounts (40), volunteered	47.9
2	What visual style, aesthetic direction, and overall branding are you envisioning?	An Elegant Traditional French look ... with a clean, modern interface	design_style (75)	55.5
3	Do you have all the content and copy ready, or will we need to create it?	The content and copy will need to be created.	content_ready (70)	62.7
4	Do you have an existing logo, specific brand colors, or any brand guidelines?	Logo: a stylized “L” ...; primary: a warm cream (#F5EFE6) ...	existing_branding (65)	69.3
5	What are your preferences for the e-commerce platform or technology stack?	We prefer WooCommerce ... a dedicated mobile app is not required.	technical_platform (60)	75.5
6	What is your estimated budget range for this project?	\$15,000 – \$35,000.	budget_range (55)	81.1
7	What is your desired launch date or timeline?	October 15th, 2024.	timeline (50)	86.2
8	Do you have any specific SEO goals, domain preferences, or search visibility strategies?	Rank highly for local searches like “French bakery Paris” ...	seo_requirements (45)	90.8

Turn 0: C0 pre-extracts purpose (100), audience (95), features (85), integrations (58): cov = 34.6. *Stop at turn 8:* cov(F) = 0.908 $\geq \theta$; unresolved content_management (35), analytics_needs (30), multilingual_support (15), accessibility (10).

Prompt P12 of the evaluation set (“I want an e-commerce site for a French bakery. It should support online ordering with delivery and pickup options, a loyalty points system, and Stripe for payments”) produces the trace of Table 1 of the main paper, which replays from `results_ALI_v1.json` in the code and data release. C0 pre-extracts four elements from the prompt alone (`main_content_purpose`, $w = 100$; `target_audience`, 95; `core_features`, 85; `integrations`, 58), so the interview opens at $338/978 \approx 35\%$ rather than at zero. C1 tracks the thirteen elements still unresolved and C2 groups them into semantic clusters. The first question resolves two of them at once (`pages_structure`, 90, and `user_accounts`, 40; +13.3 pp), and each of the seven that follow resolves one element, in strictly descending weight order (75, 70, 65, 60, 55, 50, 45): the priority mechanism, visible end to end. Coverage reaches $888/978 = 90.8\%$ at turn 8, where the loop halts on $\text{cov}(F) \geq \theta = 0.90$, having stood at 86.2% one turn earlier. On the same prompt, Claude Opus 4.8 (`results_claude-opus-4-8_multiturn_full.json`) asks six questions and declares [DONE] with eight of seventeen parameters resolved, one mentioned and eight absent, for a weighted coverage of 51.4%.

E.3 FinAgent LoRA Configuration and Training

Component names below follow the C0–C4 decomposition of Section 3 of the main paper; the implementation’s internal identifiers differ. Two of the five components are fine-tuned. The element identifier (C1) uses $r = 32$, $\alpha = 64$, targeting attention projections only (4.33 M trainable parameters). The answer extractor (C4) uses $r = 16$, $\alpha = 32$ across seven target modules including MLP projections (18.46 M trainable parameters, read from the `safetensors` headers of the shipped checkpoints), providing richer capacity for multi-parameter utterances, negation/limitation disambiguation and mid-conversation corrections. Training data: 300 samples per domain, five domains (1,500 total; 1,350 training, 150 evaluation); fixed random seed 42.

Table E.4: FinAgent training results, recomputed from the `training_history.json` of each shipped checkpoint.

Component	Base model	Best eval loss	Epochs	Best at	Halted by
C1 (element identifier)	Qwen2.5-0.5B	0.0051	6 / 30	Epoch 6	Epoch budget
C4 (answer extractor)	Qwen2.5-1.5B	0.0166	9 / 30	Epoch 4	Patience (5)

The table has two rows because FinAgent fine-tunes two components, not three: C3 generates questions through the hosted model in both deployments and is never trained, so no loss exists to report for it. Both evaluation losses are below 0.02, but they halt for different reasons and only one is a convergence result. C4 stopped on the patience criterion at a real minimum: its best epoch is the fourth of nine and every later epoch is worse (0.0236 by epoch 7). C1 stopped because its epoch budget ran out: its best epoch is its *last*, so it was still improving when training ended, and 0.0051 is an upper bound on what that configuration reaches rather than a converged value. That a loss is low says nothing about downstream usefulness in any case: the same C4 scores $\kappa = 0.0$ on structured extraction, which is why ALI_v1 keeps the API fallback.

Reproduce: the full trace is record P12 of `results/ali/results_ALI_v1.json`, scored against `data/taxonomies.json`; the training table recomputes from `training/c1/training_history.json` and `training/c4/training_history.json`.

F Question Selection: Thresholds, Reward Weights and Training Pipeline

Design choices the ablation of Section 4.4 of the main paper finds equivalent to simpler ones. This section records them for reproducibility, not as measured gains; table F.1 gives the paired ablation in full.

Table F.1: Paired ablation on $n = 30$ prompts (10 per domain): weighted coverage and turns, with 95% bootstrap CIs.

Variant	Coverage [95% CI]	Turns [95% CI]
Full pipeline	90.3 [88.6–91.9]	6.33 [5.23–7.53]
No clustering (no C2)	89.9 [88.2–91.5]	6.40 [5.27–7.57]
No priority ordering	90.2 [88.5–91.7]	6.13 [5.20–7.10]
C1 random ordering	90.2 [88.6–91.6]	6.50 [5.53–7.53]
C3 tier-1 only	90.4 [88.5–92.1]	6.57 [5.50–7.63]

Algorithm F.1 ALI conversation loop

Require: User message m_0 , threshold $\theta \in (0, 1)$, max turns $T_{\max}$

Ensure: State frame F_t with $\text{cov}(F_t) \geq \theta$

```

1: C0 (keyword scorer + regex):  $m_0 \xrightarrow{\text{detect}} d \in \mathcal{D}$ 
2: C0 (pre-extractor):  $m_0, d \xrightarrow{\text{preExtract}} F_0: \mathcal{P}_d \rightarrow \mathcal{V}; \quad t \leftarrow 0$ 
3: while  $\text{cov}(F_t) < \theta$  and  $t < T_{\max}$  do
4:   C1 (fine-tuned LM + fuzzy match):  $m_0, d, F_t \xrightarrow{\text{identify}} \mathcal{E}_t = \{(p_i, w_i)\}$  uncovered
5:   C2 (sentence-transformer + K-Means):  $\mathcal{E}_t \xrightarrow{\text{cluster}} \mathcal{C}$ , sorted by  $\sum_{p \in \mathcal{C}} w(p) \downarrow$ 
6:   C3 (3-tier: API-LLM / local LM / template):  $\mathcal{C}, \mathcal{H}_t \xrightarrow{\text{generate}} Q, |Q| = 6$ 
7:    $q^* \leftarrow \arg \max_{q \in Q} R(q, t)$  ▷ Eq. (3) of the main paper
8:   Present  $q^*$  to user; receive response  $r_t$ 
9:   C4 (fine-tuned LM + regex fallback):  $r_t, q^*, \mathcal{H}_t \xrightarrow{\text{extract}} \Delta F$ 
10:   $F_{t+1} \leftarrow F_t \cup \Delta F; \quad \mathcal{H}_{t+1} \leftarrow \mathcal{H}_t \cup \{(q^*, r_t)\}; \quad t \leftarrow t + 1$ 
11: end while
12: return  $F_t$ 

```

F.1 Dynamic Thresholds

Coverage thresholds scale with domain complexity $|\mathcal{P}_d|$ (table F.2). We determine them by Monte Carlo simulation across four questioning strategies and select the best strategy per complexity tier by

$$M_{\text{sel}} = \text{pct}_{\geq 0.90} \times 100 + \overline{\text{cov}} \times 50 - \overline{T} \times 5, \quad (\text{F.1})$$

that is, a hand-set scalarisation rewarding the share of simulated interviews that reach 90% coverage, the mean coverage and (negatively) the mean number of turns.

Table F.2: Simulation-derived starting points by domain complexity. Deployed thresholds may depart from these when weight mass is concentrated (see the FinAgent instantiation, Section 3.4 of the main paper).

Complexity	$ \mathcal{P}_d $	θ	Best strategy
Simple	≤ 10	0.75–0.85	Cluster-and-score
Medium	11–20	0.90	Cluster-and-score
Complex	21–30	0.92	Balanced
Enterprise	> 30	0.95	Balanced

F.2 Evolutionary Reward Weight Optimisation

A genetic algorithm (GA) [Holland, 1975] optimises the weight vector $\mathbf{w}$ of Eq. (3) of the main paper over simulated conversations. Population: 20 configurations, 30 generations, elite top-5; crossover uniform; mutation Gaussian ($\sim 30\%$ rate); 30% fresh injection per generation. Fitness over 200 Monte Carlo simulated conversations:

$$F(\mathbf{w}) = \overline{\text{cov}} \times 100 - \overline{T} \times 3 + (10 - \sigma_{\text{cov}}^2 \times 100)^+ + \text{pct}_{\geq 0.90} \times 20 + \frac{\overline{\text{cov}}}{\overline{T}} \times 50 \quad (\text{F.2})$$

Equation (F.2) scores a weight vector over a whole population of simulated interviews, where Eq. (F.1) scores a questioning strategy at a single complexity tier; both are hand-set scalarisations of the same three quantities (mean coverage, mean turns, share of interviews above 90%), the fitness adding a term that penalises dispersion of coverage across interviews and one that rewards coverage per turn. We report them as design choices rather than as calibrated objectives. Total budget: $30 \times 20 \times 200 = 120,000$ simulated episodes, against $\approx 2,000,000$ for a grid search. We rejected a full PPO policy: $\approx 1,100$ labelled examples is far below PPO stability thresholds, and the evolved vector can be read directly.

F.3 Seven-Phase Training Pipeline

The full ALI system trains in seven phases within ≈ 20 minutes on consumer hardware, making iterative domain adaptation practical.

- Ph1.** Generate corpus: 200 simulated episodes per domain category + 500 multi-task episodes, across four questioning strategies.
- Ph2.** Run evolutionary reward weight optimisation (Section F.2). Fresh random individuals are injected at 30% of every population to prevent premature convergence, while elitist survival keeps the best individual monotone; the generation budget is fixed in advance rather than set by a convergence test.
- Ph3.** Evaluate template effectiveness over 100 simulated conversations; compute effectiveness score = `avg_score_gained` + `bonus_extraction_rate` $\times$ 20.
- Ph4.** Optimise clusterer: test K-Means $k \in \{2, \dots, 7\}$ and agglomerative clustering; select winner by silhouette score (`n_init=50`).
- Ph5.** Run 1,000 Monte Carlo simulations per strategy; select best per complexity level by M_{sel} (Eq. (F.1)).

Ph6. Build tiered template bank: high-effectiveness elements (> 70) get assertive templates; medium (40–70) get open-ended; low (< 40) get casual phrasing.

Ph7. Validate: assert coverage $\geq 80\%$ in ≤ 8 turns. If budget remains, repeat Phases 2–3.

Reproduce: the paired ablation is the section *Ablation* ($n=30$, *paired*) of `python verify_numbers.py`; `python code/ali/ablation_stats.py` gives the Wilcoxon and TOST analysis (needs `scipy`); the records are `results/ali/results_ablation_*.json`.

G Computing Infrastructure

Every result in this paper was produced on a single consumer laptop with no discrete GPU: an Apple M4 (10 CPU cores, 16 GB unified memory) running macOS 26.5. The Telos GPT-2 and FinAgent Qwen2.5 adapters fine-tune on CPU in ≈ 15 minutes per component, and the seven-phase pipeline of the preceding appendix completes in ≈ 20 minutes on that machine. One exception: Rad-Assist’s Mistral 7B GRPO fine-tune ran on a single NVIDIA T4 GPU via Google Colab (≈ 8 hours for 1,000 reports) before being quantised to GGUF Q4_K_M for local CPU inference. The evaluated systems that are not run locally (Gemini 2.5 Flash and the three Claude models) are hosted APIs, so their inference hardware is not ours to report.

Software: Python 3.14 with `numpy` 2.4, `scipy` 1.17 (bootstrap intervals and Wilcoxon tests) and `pandas` 3.0 for analysis; `scikit-learn` 1.9 for the C2 K-Means step over embeddings from `sentence-transformers` 5.6 (all-MiniLM-L6-v2); `transformers` 5.2 with `peft` 0.18 [Mangrulkar et al., 2022] and `torch` 2.10 for fine-tuning; Ollama 0.32 for local `gemma3:4b` and `llama3.1:8b` inference; and the vendor Python SDKs for the hosted models. Wall-clock figures reported elsewhere in this document depend on API load as well as on this hardware and should not be read as hardware benchmarks.

H Real-User Validation Protocol

Real-user validation studies are *planned* for Rad-Assist and Telos; neither has collected analysable data at the time of submission, and this paper reports no result from either. We pre-register both protocols here to fix the measures and hypotheses before any data is seen.

Rad-Assist. A held-out set of 50 de-identified MIMIC-CXR reports (fixed seed) forms the evaluation set; ≥ 20 radiation oncologists dictate patient cases, and a second oncologist rates each resulting frame for completeness and clinical accuracy. Measures: weighted coverage $\text{cov}(F)$, turns to threshold, extraction F1, reviewer-rated accuracy (1–5), and the fraction of critical parameters ($w \geq 80$) resolved. The one measurement already taken on those 50 reports is output format conformity, 44/50 (88%) parsing as valid clinical frames, the six failures being prose drift on intensive-care reports (3), malformed JSON (2) and one timeout: a format check, not an accuracy result.

Telos. Twenty software developers with a concrete project each describe it, converse with Telos to $\text{cov}(F) \geq \theta$, then submit the same description to Gemini 2.5 Flash (single-pass) for within-study comparison. Hypotheses: (i) ALI turns-to-threshold $<$ naive sequential (50% reduction predicted); (ii) ALI final coverage $>$ Gemini single-pass; (iii) mean satisfaction $\geq 4.0/5.0$.

I Theoretical Analysis of the Coverage Criterion

This section analyses three structural properties of the weighted-coverage objective: its relationship to residual-entropy reduction, the sensitivity of its concentration to inter-parameter correlation, and the conditions under which greedy question selection carries a formal approximation guarantee. Two of the three results are negative; we report them because they delimit what the coverage criterion does and does not provide. We use none of them to support the empirical results of Section 4 of the main paper.

Setup. At turn t , partition the unresolved parameters into semantic clusters $C_1, \dots, C_k$ (the unit at which C2 generates questions). For cluster C_ℓ , let $W_\ell = \sum_{p_i \in C_\ell} w_d(p_i)$ be its aggregate

weight, $X_i \in \{0, 1\}$ the resolution indicator of parameter p_i with marginal π_i , and $Y_\ell = (X_i)_{i \in C_\ell}$ its joint resolution vector, with intra-cluster correlation unconstrained. We idealise cluster probes as *atomic*: a probe q_ℓ resolves all of C_ℓ with probability p_ℓ and nothing otherwise, and $\hat{H}_\ell = h(p_\ell)$ denotes the binary entropy of that outcome. We assume this idealisation throughout; the deployed question generator does not satisfy it exactly.

Lemma 1 (Entropy sub-additivity). $H(Y_\ell) \leq \sum_{i \in C_\ell} h(\pi_i)$, with equality iff the X_i are mutually independent.

Proof. By the chain rule, $H(Y_\ell) = \sum_i H(X_i | X_{<i})$, and conditioning cannot increase entropy, so $H(X_i | X_{<i}) \leq H(X_i) = h(\pi_i)$ [Cover and Thomas, 2006]. Equality in every term requires X_i independent of $X_{<i}$ for each i , i.e. mutual independence. $\square$

Consequently, any residual-entropy estimate that sums per-parameter entropies *overestimates* true residual uncertainty whenever parameters within a cluster are positively correlated. On a Medical cluster {tumour location, metastasis site} with marginals 0.6 and joint $P(1, 1) = 0.45$, the gap is $I(X_i; X_j) = 0.1024$ bits, i.e. 5.3% of the summed entropies.

Theorem 1 (Coverage–entropy comonotonicity). *Assume every cluster admits an atomic probe in the question pool. The coverage-gain ranking Φ_W and the entropy-reduction ranking $\Phi_{\hat{H}}$ induce the same preorder over probes if and only if the expected gains $(p_\ell W_\ell)$ and the residual entropies $\hat{H}$ are comonotone across clusters, that is $p_\ell W_\ell > p_{\ell'} W_{\ell'} \Leftrightarrow \hat{H}_\ell > \hat{H}_{\ell'}$ for all ℓ, ℓ' . Comonotonicity of the curated weights W with $\hat{H}$ is neither necessary nor sufficient for this.*

Proof. Under atomicity, probing q_ℓ yields expected coverage gain $\Phi_W(q_\ell) = p_\ell W_\ell$ and expected entropy reduction $\Phi_{\hat{H}}(q_\ell) = h(p_\ell) = \hat{H}_\ell$. Two real-valued scores induce the same preorder exactly when they order every pair of probes alike, which is the stated condition on $(p_\ell W_\ell)$ and $\hat{H}$. That the condition cannot be transferred to W follows from a counterexample: with $W_1=0.45$, $p_1=0.90$ and $W_2=0.55$, $p_2=0.50$ we have $W_2 > W_1$ and $\hat{H}_2=1.0000 > \hat{H}_1=0.4690$, so W and $\hat{H}$ are comonotone, yet $\Phi_W(q_1) = 0.405 > \Phi_W(q_2) = 0.275$ inverts the entropy ranking. The resolution probability p_ℓ enters the two scores differently and is not monotone in W_ℓ . $\square$

Comonotonicity is a substantive condition that ALI’s design does not enforce: domain weights w_d are fixed by taxonomy curation, without reference to the empirical p_ℓ . It fails in realistic configurations. Taking the cluster above as C_1 ($W_1=0.45$, $p_1=0.45$) against the rest of the domain as C_2 ($W_2=0.55$, $p_2=0.70$), the two rankings invert: $\hat{H}_1=0.9928 > \hat{H}_2=0.8813$ while the expected coverage gains order the probes the other way ($p_2 W_2=0.385 > p_1 W_1=0.2025$). Coverage-driven selection is therefore *not* entropy-optimal in general.

Theorem 2 (Correlation sensitivity of the stopping guarantee). *Let $\text{cov} = \sum_i w_i X_i$ with $\sum_i w_i = 1$. Then (a) $\mathbb{E}[\text{cov}] = \sum_i w_i \pi_i$ for any joint law of the X_i with fixed marginals; (b) $\text{Var}(\text{cov}) = \sum_i w_i^2 \pi_i (1 - \pi_i) + \sum_{i \neq j} w_i w_j \text{Cov}(X_i, X_j)$, and for every $\rho \in [0, 1]$ there exists a joint law with the prescribed marginals attaining $\text{Cov}(X_i, X_j) = \rho \pi_i (1 - \pi_j)$; (c) $P(\text{cov} \leq \mu - \kappa) \leq \sigma^2 / (\sigma^2 + \kappa^2)$, where $\mu = \mathbb{E}[\text{cov}]$ and $\sigma^2 = \text{Var}(\text{cov})$.*

Proof. (a) Linearity of expectation does not involve the joint law. (b) Bilinearity of covariance gives the stated decomposition. For realisability, let Z and $U_1, \dots, U_n$ be independent Bernoulli(π) variables and, independently for each i , set $X_i = Z$ with probability $\sqrt{\rho}$ and $X_i = U_i$ otherwise. Each X_i is Bernoulli(π), and for $i \neq j$ the two copy Z simultaneously with probability ρ , giving $\text{Cov}(X_i, X_j) = \rho \pi (1 - \pi)$; every level in $[0, 1]$ is therefore attained by an admissible law. (c) Cantelli’s one-sided inequality applied to $-\text{cov}$. $\square$

Expected coverage is thus invariant to correlation, but the *guarantee* against under-coverage degrades. With $n=5$, $w_i=0.2$, $\pi_i=0.5$ and $\rho=0.3$, the variance rises from 0.05 to 0.11 ($\times 2.2$) and the Cantelli bound at $\kappa=0.4$ weakens from 23.8% to 40.7% ($\times 1.71$). These are worst-case

bounds rather than realised probabilities; in practice, positive intra-cluster correlation leaves mean coverage untouched while widening the dispersion the stopping rule must absorb.

Theorem 3 (Adaptive submodularity: refutation and restricted validity). *(a) cov is not adaptive submodular in general. (b) Under H1' (mutual independence across clusters, with intra-cluster correlation unconstrained) and atomic cluster probes, cov at cluster level is adaptive monotone and adaptive submodular, and greedy cluster selection satisfies $\mathbb{E}[\text{cov}(F_T)] \geq (1 - 1/e) \cdot \text{OPT}_T$.*

Proof. (a) Take the correlated Medical cluster above, with $w = 0.2$ for each parameter. Before any observation, the expected marginal gain of probing the metastasis parameter is $w \cdot P(X_j=1) = 0.2 \times 0.6 = 0.120$. After observing $X_i=1$, Bayes gives $P(X_j=1 \mid X_i=1) = 0.45/0.6 = 0.75$, so the expected marginal gain becomes $0.2 \times 0.75 = 0.150 > 0.120$. Adaptive submodularity requires expected marginal gains to be non-increasing as observations accumulate; this is a direct violation. (b) Under H1' the resolution vectors of distinct clusters are independent, and an atomic probe resolves its cluster wholly or not at all, so the state of one cluster carries no information about another. The problem is then an instance of adaptive stochastic maximum coverage over independent items, which is adaptive monotone and adaptive submodular; the $(1 - 1/e)$ bound for the greedy policy follows from the general result of Golovin and Krause [2011]. $\square$

H1' is strictly weaker than full parameter independence (it permits the intra-cluster facilitation that refutes the general claim) but it remains an *assumption*: we have not verified it empirically on the three deployment taxonomies, and the $(1 - 1/e)$ guarantee should be read as conditional on it. Taken together, theorem 1 and theorem 3 delimit the criterion: weighted coverage is a well-defined completeness target with a conditional greedy guarantee, and entropy optimality is not among its properties. This is consistent with the ablation result of Section 4.4 of the main paper (the question-selection refinements carry no entropy-optimality property that disabling them could degrade), but it does not predict that result, and we do not claim it as an explanation of the measured coverage gain.

J Positioning Against Prior Work

Backs contribution (1), and states its boundary. This section writes the novelty claim as a conjunction of four properties, shows that no prior system assembles the conjunction, and records how we ran the search and what would refute it, since a negative claim about a literature is only as good as its search.

Table J.1: The four properties of the claim against one representative of each neighbouring family, all discussed in the text. $\checkmark$ present, $\circ$ partial, $—$ absent. The last column is where the two families separate.

Work	A	B	C	D	Ends on
IN3 / Tell Me More!	$\circ$	$—$	$\checkmark$	$—$	A vagueness judgement
OntoAgent	$\circ$	$—$	$\circ$	$—$	An ontology, traversed
SAGE-Agent	$—$	$\circ$	$\checkmark$	$—$	A belief threshold
ReqElicitGym	$\circ$	$—$	$—$	$—$	N/A: scored offline
Slot filling, DST	$—$	$—$	$\circ$	$—$	Every slot filled
ALI (this work)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\text{cov}(F) \geq \theta_d$

What is claimed. We claim no individual mechanism: weighted parameter spaces, semantic clustering, low-rank adapters and simulated-user evaluation are all standard, and the main paper’s ablation reports that three of ALI’s own refinements contribute nothing measurable. We claim a conjunction of four properties holding at once in one running system:

(A) Weighted coverage: a continuous $\text{cov}(F) \in [0, 1]$ over a parameter space $\mathcal{P}_d$ fixed per domain and carrying heterogeneous importance weights w_d , rather than a count of filled slots or a post-hoc score. **(B) An online formal stopping rule:** that same quantity *is* the termination condition, tested against a per-domain threshold θ_d , rather than a learned judge

declaring itself satisfied, a posterior variance or a turn budget. **(C) Selection by coverage gain:** each turn asks the question maximising expected coverage gain, several parameters at a time through semantic clustering. **(D) Cross-domain validation:** the same five-component decomposition, unmodified, on three separately built deployments. Prior work occupies each property alone, and the main paper cites the work that occupies it; table J.1 is the claim in its checkable form, and the conjunction $A \wedge B \wedge C \wedge D$ has one row.

The fault line. The literature nearest to us splits into two families that do not meet. One *measures* something like coverage, and measures it offline: ReqElicitGym after a transcript exists [Jin et al., 2026b], τ -bench against a final database state [Yao et al., 2024], CLAMBER on the ambiguity an answer should have resolved [Zhang et al., 2024a], PARADISE-style evaluation once the dialogue is over [Walker et al., 1997]. In none of them does the agent read the coverage figure while it is interviewing. The other family *stops* on a criterion the agent evaluates online, but the criterion is information-theoretic: SAGE-Agent halts when the belief over candidate tool calls concentrates past a threshold or when expected value of information falls below its cost [Suri et al., 2025], BED-LLM on posterior uncertainty under a Bayesian experimental-design objective [Choudhury et al., 2026]. Both are more principled than a coverage threshold about what they measure, and neither measures what a domain owner curated as mattering. We are not aware of work that makes an importance-weighted coverage function both the objective and the calibrated stopping criterion of an active interview.

The nearest neighbours. IN3 and *Tell Me More!* hold (A, in part) and (C) but end on a trained vagueness judgement, so a reader cannot ask of a transcript what fraction of the task it settled [Qian et al., 2024]; OntoAgent ranks ontology concerns with no numeric completeness threshold and one evaluated domain, and we assess it from its abstract rather than from a reimplementaion [Jin et al., 2026a]. The rest decide question by question whether to clarify at all [Zhang et al., 2024b, Kuhn et al., 2022, Rao and Daumé III, 2018, Majumder et al., 2021, Aliannejadi et al., 2020], or supply frame-filling and turn-taking machinery without importance weights or a measurable stopping condition [Young et al., 2013, Williams et al., 2016, Henderson et al., 2014, Budzianowski et al., 2018, Allen et al., 1999]; none carries a budget over a whole parameter space [Deng et al., 2025].

What we do not claim. Four things. Not that the criterion is optimal: Section I shows it is not entropy-optimal in general. Not a guarantee of the kind adaptive submodularity provides for greedy active selection [Golovin and Krause, 2011]; we do not show that our objective satisfies its conditions. Not priority over any single component. And not an external benchmark result: we have not evaluated ALI on IN3, τ -bench or ReqElicitGym, the single most useful comparison a reader could ask for, and that comparison is future work rather than a result of this paper. The same authors wrote the evaluation instrument and the systems it evaluates.

How we ran the search, and what would refute it. We ran the survey in July 2026 over arXiv, the ACL Anthology, OpenReview, IEEE Xplore, the ACM Digital Library and Semantic Scholar, along six axes from coverage metrics for dialogue to surveys of proactive dialogue, and we read directly every candidate that threatened the claim. Two caveats: this is a broad sweep rather than a systematic review, and no sweep proves a negative; and several assessments rest on abstracts rather than reimplementaion. We therefore state the claim as *we are not aware of*, and one step refutes it: a published system halting an interview on a calibrated threshold over a weighted, domain-fixed parameter space would settle it. A system computing such coverage offline would not, and neither would one halting on uncertainty: those are the two families above, and the distinction is the contribution.

K Limitations

Section 5 of the main paper summarises the limitations in prose; this section states each in full.

Training data realism. Telos and FinAgent train entirely on programmatically generated data. We applied five augmentation strategies to cover diverse input styles, but real users produce patterns generators cannot anticipate: unusual phrasings, domain crossover queries, multilingual inputs, evasive or contradictory answers. Rad-Assist narrows this gap by grounding C1 and C4 training in real anonymised MIMIC-CXR reports [Johnson et al., 2019]; evaluation of the full pipeline with real oncologists is still in progress.

Multi-turn LLM comparison. Phase 1 provides multi-turn baselines for three accessible models (42.2–52.8%, $n = 50$); we added three frontier baselines from a single family (Claude Haiku 4.5, Sonnet 5, Opus 4.8) later. Those runs deviate from the Phase 1 protocol in documented ways: the simulated-user oracle is Claude Haiku 4.5 rather than a local 4B model, though the scorer remains Gemini and so stays in a different model family from every interviewer; no temperature is set, because the API removed the parameter for this model generation; adaptive reasoning is enabled for Opus and Sonnet but rejected by the API for Haiku; and we ran Sonnet 5 on the 50-prompt subset only, for budget reasons. These are best-effort rather than constrained configurations. Three asymmetries also favour ALI and were not controlled. The baseline interviewer was instructed to finish within five to eight questions while ALI runs to a 15-turn cap; the baseline oracle may answer “I’m not sure” to a technical question whereas ALI’s oracle is instructed to invent a plausible concrete value, which is what the scorer credits as resolved; and ALI reads the same taxonomy the scorer marks against, which no baseline sees. The truncation control in Section C bounds the first at 4.6 pp. The other two are untested, so the reported gap is an upper bound on what the stopping rule alone contributes, and the flat capability curve is measured at a fixed turn budget. Frontier models outside this family (GPT-4o class) remain unevaluated.

Static reward weights. We optimise reward weights at training time; they do not adapt to individual user behaviour. A learned policy could discover patterns once sufficient real conversation data accumulates.

Fixed taxonomy. Adapting to a new domain requires one to two days of domain expert effort. An auto-discovery mechanism based on topic modelling would reduce this cost.

Advisory enforcement in Rad-Assist and Telos. Unlike FinAgent, these deployments have no enforcement layer; downstream correctness depends on extraction accuracy, acceptable given lower irreversibility.

Error propagation. The five-component cascade creates compound failure modes. Errors in C0 corrupt C1; errors in C1 propagate to C2 and C3; C4 errors corrupt the state frame outright. We have not characterised end-to-end extraction accuracy across realistic inputs.

Reward weight justification. The weight vector $\mathbf{w}$ derives from evolutionary optimisation rather than theoretical analysis. The fitness function (Eq. (F.2)) encodes unvalidated design choices; sensitivity analysis and validation on real user data remain open tasks.

Statistical power. The single-pass baseline uses $n = 150$ prompts (50 per domain) with 95% bootstrap CI; the Phase 1 multi-turn baselines use $n = 50$ (oracle user). Those 50 prompts are a strict subset of the 150-prompt set used by ALI and by the Opus 4.8 and Haiku 4.5 baselines, so the headline comparison (ALI vs. Claude Opus 4.8) is fully paired. Within-domain variance remains high (std 13–24 pp); the Payments scorer also fails its pre-registered $\kappa \geq 0.65$ threshold, so Payments figures carry additional uncertainty.

Oracle-simulated user. Both the multi-turn LLM baselines and the ALI evaluation use a second LLM to simulate user responses. Simulated users may respond more cooperatively than real users, which would inflate coverage estimates for all systems. Real-user studies are the next planned step, now unblocked by Phase 2.

Payments taxonomy ambiguity. The LLM-as-judge scorer reaches $\kappa = 0.539$ on Payments, below the pre-registered $\kappa \geq 0.65$ threshold. Three refinement attempts (original taxonomy, regex-corrected labels, LLM re-annotation) all failed; the LLM re-annotation yielded $\kappa = 0.349$, worse than the original. A control judge from a different model family (Claude Sonnet 5, same 516

human-annotated items) is weaker than the scorer of record in *every* domain (0.529 Consulting, 0.605 Medical, 0.294 Payments), so it does not by itself corroborate a Payments-specific defect. What it does show is narrower and still informative: Payments is the worst domain for both judges, while the two judges agree with *each other* about equally in all three (0.589–0.632). The root cause is directional disagreement: human annotators accept vaguer mentions as RESOLVED where both scorers require explicit quantitative values. We therefore report this as a property of the Payments taxonomy rather than as a scorer defect, and note that further label refinement is not the remedy: the parameter space itself would need redesign, with resolution criteria fixed operationally before annotation rather than inferred from it. The limitation is also conservative with respect to our main claim: Payments is Claude Opus 4.8’s *strongest* domain (63.6%) and ALI_v1’s *lowest-scoring* one (83.9%, at its own lower threshold $\theta = 0.75$), so dropping it *widens* rather than narrows the gap. On Consulting and Medical alone ($n = 100$ paired prompts), Opus 4.8 reaches 55.8% [52.6–59.0%] against 91.6% [90.8–92.3%] for ALI_v1, a gap of +35.8 pp against +30.6 pp over all three domains. The Payments figures in the main paper should nonetheless be read with this caveat in mind.

Fine-tuned C4 at the scales we tested. C4, fine-tuned on GPT-2 124M (Telos) and Qwen2.5-1.5B (FinAgent), fails to produce the structured output format the pipeline requires ($\kappa = 0.0$ for both deployments). This limits ALI_v3 to near-baseline performance (51.6%) and makes fine-tuned C4 non-viable at these scales. The fallback to Gemini 2.5 Flash for C4 (ALI_v1) therefore remains necessary across the whole 0.1–1.5B range we fine-tuned; we have not tested where, above it, the fallback stops being needed.

C1 recall measurement. ALI_v1 C1 diagnostics report recall = 1.0 for Telos and FinAgent, which is a structural artefact: when fine-tuned C1 fails, the pipeline falls back to the full unresolved parameter list. The metric therefore reflects the fallback rather than true C1 quality. We defer a proper ablation isolating C1.

Ablation scope. The component ablation of Section 4.4 of the main paper uses a paired subset of $n = 30$ prompts, powered to detect coverage effects of 0.8–1.5 pp. Equivalence is established within margins of 2 pp and one turn, so the absence of effect is demonstrated rather than unrejected. Two caveats remain. First, equivalence holds *within those margins*: effects smaller than 2 pp are not excluded; they are declared irrelevant at the scale of the reported coverage gap. Second, each variant degrades a single mechanism in isolation; the ablations do not test whether the components are jointly redundant, and they leave the coverage bookkeeping and the stopping rule intact throughout.

Reproduce: the figures quoted in this section recompute with `python verify_numbers.py`; the equivalence test with `python code/ali/ablation_stats.py`.

L Ethics Statement

Two ethical considerations arise from the deployments described in this paper. First, Rad-Assist uses data from MIMIC-CXR [Johnson et al., 2019], a dataset of de-identified radiology reports released under a PhysioNet Credentialed Health Data License; we collected and processed no additional patient data. Second, FinAgent is designed for autonomous agents executing real financial transactions. The weighted coverage metric provides a measurable indicator of constraint completeness but is not a sufficient safety guarantee on its own; expert review and formal validation of the enforcement engine remain necessary before any production deployment. The adversarial test suite behind that layer spans seven attack categories and $\approx 124k$ generated cases; the validator raised no exception on any of them, including hostile input types. That result is an internal-consistency check and nothing more: the expected decisions are computed by functions mirroring the validator’s own specification, so full agreement shows that the implementation matches its specification, not that the specification is right. It establishes neither coverage of real-world attack scenarios nor completeness of the specification itself, and it says nothing about the intent-understanding accuracy that is this paper’s result.

References

- Mohammad Aliannejadi, Julia Kiseleva, Aleksandr Chuklin, Jeff Dalton, and Mikhail Burtsev. ConvAI3: Generating clarifying questions for open-domain dialogue systems (ClariQ). *arXiv preprint arXiv:2009.11352*, 2020. URL <https://arxiv.org/abs/2009.11352>.
- James E. Allen, Curry I. Guinn, and Eric Horvitz. Mixed-initiative interaction. *IEEE Intelligent Systems and their Applications*, 14(5):14–23, 1999. doi: 10.1109/5254.796083.
- Paweł Budzianowski, Tsung-Hsien Wen, Bo-Hsiang Tseng, Iñigo Casanueva, Stefan Ultes, Osman Ramadan, and Milica Gašić. MultiWOZ – A large-scale multi-domain wizard-of-Oz dataset for task-oriented dialogue modelling. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018. doi: 10.18653/v1/D18-1547.
- Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, Yizhe Zhang, and Tom Rainforth. BED-LLM: Intelligent information gathering with LLMs and Bayesian experimental design. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. URL <https://arxiv.org/abs/2508.21184>.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- Yang Deng, Lizi Liao, Wenqiang Lei, Grace Hui Yang, Wai Lam, and Tat-Seng Chua. Proactive conversational AI: A comprehensive survey of advancements and opportunities. *ACM Transactions on Information Systems*, 43(3), 2025. doi: 10.1145/3715097. URL <https://dl.acm.org/doi/10.1145/3715097>.
- Daniel Golovin and Andreas Krause. Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *Journal of Artificial Intelligence Research*, 42:427–486, 2011. doi: 10.1613/jair.3278.
- Matthew Henderson, Blaise Thomson, and Jason D. Williams. The second dialog state tracking challenge. In *Proceedings of the 15th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)*, 2014. doi: 10.3115/v1/W14-4337.
- John Henry Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, Ann Arbor, MI, 1975.
- Dongming Jin, Zhi Jin, Yaotian Yang, Linyu Li, Zheng Fang, Yuanpeng He, Wenchun Jing, and Xiaohong Chen. From chat to interview: Agentic requirements elicitation with an experience ontology. *arXiv preprint arXiv:2605.05828*, 2026a. URL <https://arxiv.org/abs/2605.05828>.
- Dongming Jin et al. ReqElicitGym: An evaluation environment for interview competence in conversational requirements elicitation. *arXiv preprint arXiv:2602.18306*, 2026b. URL <https://arxiv.org/abs/2602.18306>.
- Alistair E. W. Johnson, Tom J. Pollard, Seth J. Berkowitz, Nathaniel R. Greenbaum, Matthew P. Lungren, Chih-ying Deng, Roger G. Mark, and Steven Horng. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data*, 6:317, 2019. doi: 10.1038/s41597-019-0322-0.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. CLAM: Selective clarification for ambiguous questions with generative language models. *arXiv preprint arXiv:2212.07769*, 2022. URL <https://arxiv.org/abs/2212.07769>.

- Bodhisattwa Prasad Majumder, Sudha Rao, Michel Galley, and Julian McAuley. Ask what’s missing and what’s useful: Improving clarification question generation using global knowledge. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 4300–4312, 2021. doi: 10.18653/v1/2021.naacl-main.340.
- Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, Younes Belkada, Sayak Paul, and Benjamin Bossan. PEFT: State-of-the-art parameter-efficient fine-tuning methods, 2022. URL <https://github.com/huggingface/peft>.
- Cheng Qian, Bingxiang He, Zhong Zhuang, Jia Deng, Yujia Qin, Xin Cong, Zhong Zhang, Jie Zhou, Yankai Lin, Zhiyuan Liu, and Maosong Sun. Tell me more! Towards implicit user intention understanding of language model driven agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. URL <https://aclanthology.org/2024.acl-long.61/>.
- Sudha Rao and Hal Daumé III. Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018. doi: 10.18653/v1/P18-1255.
- Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, and Dinesh Manocha. Structured uncertainty guided clarification for LLM agents. *arXiv preprint arXiv:2511.08798*, 2025. URL <https://arxiv.org/abs/2511.08798>.
- Marilyn A. Walker, Diane J. Litman, Candace A. Kamm, and Alicia Abella. PARADISE: A framework for evaluating spoken dialogue agents. In *Proceedings of the 35th Annual Meeting of the Association for Computational Linguistics (ACL)*, 1997. doi: 10.3115/976909.979652.
- Jason D. Williams, Antoine Raux, and Matthew Henderson. The dialog state tracking challenge series: A review. *Dialogue and Discourse*, 7(3):4–33, 2016. doi: 10.5087/dad.2016.302.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024. URL <https://arxiv.org/abs/2406.12045>.
- Steve Young, Milica Gašić, Blaise Thomson, and Jason D. Williams. POMDP-based statistical spoken dialogue systems: A review. *Proceedings of the IEEE*, 101(5):1160–1179, 2013. doi: 10.1109/JPROC.2012.2225812.
- Tong Zhang, Peixin Qin, Yang Deng, Chen Huang, Wenqiang Lei, Junhong Liu, Dingnan Jin, Hongru Liang, and Tat-Seng Chua. CLAMBER: A benchmark of identifying and clarifying ambiguous information needs in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10746–10766, 2024a. doi: 10.18653/v1/2024.acl-long.578. URL <https://aclanthology.org/2024.acl-long.578/>.
- Xuan Zhang, Yang Deng, Zifeng Ren, See-Kiong Ng, and Tat-Seng Chua. Ask-before-Plan: Proactive language agents for real-world planning. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024b. URL <https://aclanthology.org/2024.findings-emnlp.636/>.